

PROGRESS AND INEQUALITY

COURSE ECON 30

TERM Spring 2026

CLASS Tue & Thu 9:00–10:20am, Lathrop 299

INSTRUCTOR [Lukas Althoff](#)

TEACHING ASSISTANT TBD

COURSE DESCRIPTION Progress and inequality are intertwined throughout human history. This course traces both from the Neolithic Revolution to the AI era, asking: What causes progress? Who benefits? And what can we learn for the future? Topics range from the engines of progress—resources, cities, knowledge, institutions—to the barriers that determine who is left behind, including race, gender, and intergenerational mobility.

COURSE FORMAT This course consists of two weekly classes. Assignments may include readings, writing short papers, using AI tools to design interactive data visualizations, peer review, and group projects.

LEARNING GOALS You will (1) understand the long-run evolution of progress and inequality around the world and the US, (2) gain competency in using generative AI, coding, and data analysis tools to combine different media to tell compelling data-driven stories about the societal challenges, and (3) evaluate causal claims based on quantitative evidence and quantitative economic methods.

REQUIREMENTS No prior coursework in economics required. You are required to (1) use advanced generative AI tools (e.g., CursorAI, Claude Code), (2) be comfortable working with at least one coding language (e.g., Python, R, Stata, Julia, C++, etc.), and (3) be able to independently debug and problem-shoot code. Active participation in class and thorough preparation of and reflection on readings in advance is required.

INTERACTIVE COMPONENTS & REQUIREMENTS

1. [Assignments](#) (20% of grade)

Weekly assignments that build toward the capstone project, including peer evaluation.

2. **Class participation** (30% of grade)

Attendance (mandatory) and quality of contributions to class discussion. Asking good questions counts as much as answering them.

3. **Capstone draft** (10% of grade)

A mid-quarter draft of the capstone project, submitted for peer feedback.

4. **Capstone project** (30% of grade)

An original research project on any aspect of progress that you are excited about. You present your findings through a mix of quantitative analysis, data visualization, and narrative exposition.

AI POLICY You are highly encouraged to make ethical use of AI tools (e.g., exploring non-required readings through AI-generated podcasts or summaries, interact with readings through AI-powered chats, generate data using AI tools, AI code assistance, etc.). Of course, you remain responsible for the content of your work including accuracy, originality, writing style, etc. (“garbage in, garbage out”). Grade bonuses are awarded for exceptionally creative use of AI tools, such as collecting new data with the help of AI or creating an original public good relevant to this course.

NO-SCREEN POLICY Please do not use your phone, iPad, or laptop during class, unless prompted by the instructor.

COURSE SCHEDULE

Part I: Long-run economic progress and inequality

1. THE MIRACLE OF THE MODERN WORLD

The extraordinary progress of the last few centuries in historical context. Life before modern technology. The hockey-stick of human development.

2. THE NEOLITHIC REVOLUTION — WHEN INEQUALITY BEGAN

Why did agriculture emerge simultaneously across the globe? Climate variability, surplus, hierarchy, and the birth of inequality.

3. THE LONG STAGNATION AND THE INDUSTRIAL ESCAPE

The Malthusian trap: why growth was near-zero for millennia. Why Britain? Why the 18th century? The invention of the method of invention.

Part II: Engines of Progress

4. RESOURCES AND CLIMATE

From fire to fossil fuels to renewables. Materials from stone to silicon. Climate change as the cost of fossil-fueled progress: damages six times larger than previously estimated (Bilal & Känzig). Energy poverty, inequality, and the transition dilemma.

5. TRIUMPH OF THE CITY

Agglomeration, knowledge spillovers, and innovation clusters. Spatial inequality and housing constraints.

6. COLLECTIVE INTELLIGENCE — HUMANITY'S CUMULATIVE ADVANTAGE

Oral tradition to printing press to internet to AI. The science of science. Are ideas getting harder to find—or is the real bottleneck diffusion?

7. INSTITUTIONS — THE RULES OF THE GAME

Inclusive vs. extractive institutions. Colonial origins of comparative development. The 2024 Nobel Prize framework.

8. ORGANIZATIONS — WHO GETS TO BUILD?

Bell Labs, Skunk Works, DARPA. Why some organizations produce breakthroughs and who gets access.

9. INCENTIVES — WHAT WE CHOOSE TO SOLVE

Patents, prizes, grants, and markets. Innovation follows purchasing power, not need. The 90/10 gap. Inappropriate technology for the developing world (Moscona & Sastry).

10. TALENT AND AMBITION

Lost Einsteins and the allocation of talent. Suppressed ambition as lost progress.

Part III: Inequality as the Shadow of Progress

11. GLOBAL INEQUALITY

Why some nations are rich and others poor. Colonialism as extraction. Globalization's double edge. Can the global periphery catch up?

12. WITHIN-COUNTRY INEQUALITY

The classical U-shape of inequality (Piketty). Skill-biased technological change, taxation, and the debate over how much inequality has really increased (Auten & Splinter vs. Piketty & Saez).

13. RACIAL INEQUALITY

Slavery, Jim Crow, redlining, and the persistence of racial inequality through ostensibly race-neutral institutions.

14. GENDER INEQUALITY

The U-shaped history of women's labor force participation. The child penalty. Progress for whom?

15. INTERGENERATIONAL MOBILITY

What determines mobility across generations? Mothers, schools, colleges as mobility engines. The broken ladder and lost potential.

Part IV: The Future of Progress

16. THE AI ERA — GROWTH, INEQUALITY, AND RISK

Can AI accelerate progress? Who wins and who loses? The optimist, skeptic, and "real bottlenecks" cases. Existential risk and the energy constraint.

17. STUDENT PRESENTATIONS I

Students present their capstone projects and receive feedback from peers and instructor.

18. STUDENT PRESENTATIONS II

Continued student presentations. Reflections on common threads and tensions across projects.

19. BUILDING THE FUTURE

Progress studies, degrowth, effective altruism, solarpunk. What kind of future do we want to build?